

Paper III - Subspeciality of Urology (practice paper, MCh Urology Summer 2015 set)

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III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Type: LAQ (25 marks)

Introduction / History

- A hollow tube that keeps the upper tract draining past obstruction, injury or anastomosis.
- **Zimskind 1967**: first indwelling silicone splint; being straight, it migrated.
- **Finney 1978**: the **double-J** stent. Self-retaining coils solved migration.

Stent design

Parameter	Details
Diameter	3–8.5 Fr (adults 4.7–7 Fr, usually 6 Fr). Bigger stents drain better in extrinsic obstruction because most flow is around the stent
Length	Adults 20–30 cm (usually 24–26 cm), chosen by height; children about age + 10 cm
Flow	Poor in malignant compression; use tandem or metallic stents or PCN (percutaneous nephrostomy)

Diagram to draw: double-J stent in situ with renal coil, side holes and bladder coil.

Stent materials

Material	Features
Silicone	Most biocompatible, least encrustation, but soft and hard to push
Polyurethane	Stiff and cheap, but more encrustation and irritation
Metallic (Resonance)	Resists compression in malignant obstruction; licensed for up to 12 months
Memokath 051	Thermo-expandable nitinol for long-term strictures
Coated	Heparin (anti-encrustation), triclosan (no convincing benefit), drug-eluting (investigational)
Biodegradable	No removal needed, so no forgotten stent; still experimental because of fragmentation

Special stents

- Magnetic-tip stents and stents on a string can be removed without cystoscopy.

Indications

- Relief of obstruction: stone with sepsis (stent is as good as PCN), malignant obstruction, stricture, pregnancy.
- After intervention: URS (ureteroscopy) in selected cases, pyeloplasty, reimplantation, transplant.
- To help healing of injury, leak or fistula, and prophylactically before complex pelvic surgery.

Stent-related symptoms

- Up to about 80% have frequency, urgency, flank pain on voiding or haematuria, measured by the validated **USSQ (Ureteral Stent Symptom Questionnaire)**.
- Causes: trigonal irritation when the bladder coil crosses the midline, reflux of urine up the stent during voiding, and ureteric spasm.
- Treat with an **alpha-blocker plus an antimuscarinic**; the combination beats either alone. Use the correct length and shortest dwell.

Complications

- Migration: a stent that is too short migrates upwards and needs URS retrieval.
- Infection: colonisation is near-universal with longer dwell.
- **Encrustation** increases with time (risk rises after 6 weeks–3 months), especially in stone formers, urease-producing UTI (urinary tract infection) and CKD (chronic kidney disease).
- A **forgotten stent** can cause obstruction, sepsis and loss of the kidney; registries prevent this.
- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear; II circular; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.

Management of encrusted stent

- Never pull it out forcefully, because of the risk of avulsion. Assess the stone burden with NCCT (non-contrast computed tomography) and treat infection first.
- Clear the **bladder coil** first (cystolithotripsy), then the **ureter** (URS with laser), then the **renal coil** (flexible URS, or PCNL (percutaneous nephrolithotomy) for a large burden).
- One session where possible; nephrectomy if non-functioning.

Ureteroscopy

	Semirigid	Flexible
Size	~6–8.5 Fr	~7.5–9.5 Fr
Best for	Distal and mid ureter	Upper ureter and kidney
Deflection	None	Up to ~270° in both directions

- **UAS (ureteral access sheath)**, 10/12–12/14 Fr, lowers intrarenal pressure and allows repeated passage. It can injure the ureteric wall (about 1 in 8 significant); pre-stenting passively dilates the ureter and reduces this.
- **Ho:YAG (holmium:YAG) laser, 2120 nm**, is the gold standard. **TFL (thulium fibre laser), ~1940 nm**, gives finer dust and less retropulsion.
- Dusting is faster but leaves fine fragments. Fragmentation with basketing confirms stone-free status and gives stone for analysis.
- Steps: safety guidewire, semirigid inspection, access sheath, laser and basket, inspect the ureter on withdrawal, decide on a stent.

Stenting after URS

- **AUA (American Urological Association) 2016:** a stent may be omitted after uncomplicated URS.
- **EAU (European Association of Urology) 2024:** do not routinely stent after uncomplicated URS.
- Omitting the stent gives less pain and fewer LUTS (lower urinary tract symptoms) at lower cost, without more unplanned visits.
- **Stent is indicated for:** ureteric injury or perforation, significant residual fragments, solitary or transplant kidney, impacted stone or oedema, pregnancy, infection, renal insufficiency, bilateral URS.
- Duration: 3–7 days (often on a string) when uncomplicated; 2–6 weeks after injury.

After URS -> uncomplicated?

Yes -> no stent (or string stent 3-7 d)

No -> double-J: injury 2-6 wk; residual/impacted 1-2 wk

Recent advances

- **FANS (flexible and navigable suction access sheath):** lower intrarenal pressure, better stone-free rate, fewer infections.
- Single-use digital flexible ureteroscopes.

Key points to remember:

- Finney 1978: double-J stent.
 - Alpha-blocker plus antimuscarinic best relieves stent symptoms.
 - Encrusted stent: clear bladder coil, then ureter, then kidney.
 - No routine stent after uncomplicated URS (AUA 2016, EAU 2024).
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III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR.

Type: LAQ (25 marks)

Definition

- **VUR (vesicoureteral reflux)** is backflow of urine from bladder to ureter and kidney.
- **Primary** VUR is a congenital deficiency of the UVJ (ureterovesical junction) flap-valve with a short intravesical tunnel (normal length to diameter ratio about 5:1).
- **Secondary** VUR results from high bladder pressure or obstruction: PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction).
- Found in about 1% of normal children, 30–40% of children with febrile UTI (urinary tract infection) and 30% of siblings.

Pathophysiology

- Reflux with infection causes pyelonephritis; compound papillae allow intrarenal reflux and **scarring**.
- Scarring causes hypertension, proteinuria and CKD (chronic kidney disease): **reflux nephropathy**.

IRS (International Reflux Study) grading on VCUG (voiding cystourethrogram)

Grade	Findings
I	Contrast in the ureter only
II	Ureter, pelvis and calyces, no dilatation
III	Mild-moderate dilatation, minimal blunting of fornices
IV	Moderate dilatation; fornices obliterated but papillary impressions maintained
V	Gross dilatation; papillary impressions lost

Diagram to draw: grades I–V sketches and a Cohen cross-trigonal reimplant.

Natural history

- Resolution depends on grade, laterality, age and BBD.
- Grades I–II: about 80% or more resolve over ~5 years. Grade III: about 50%.
- Grade IV: unilateral about 50–60% at 5 years, but **bilateral only about 10%**. Grade V rarely resolves.

Evaluation

- History of febrile UTI and BBD; examine BP and spine. Urine culture, creatinine, urine protein.
- **RBUS (renal-bladder ultrasound)**: hydronephrosis, renal size, post-void residual.
- **VCUG**: grade, and the urethra in boys to exclude PUV.
- **DMSA (dimercaptosuccinic acid) scan**: scars and differential renal function.

Medical management

1. CAP (continuous antibiotic prophylaxis): a single nightly dose, about a quarter of the treatment dose.

Drug	Dose	Notes
TMP-SMX (trimethoprim-sulfamethoxazole)	2 mg/kg TMP	Avoid under 2 months (kernicterus)
Nitrofurantoin	1-2 mg/kg	Avoid in CKD and G6PD deficiency
Cephalexin	~10 mg/kg	For infants under 2 months

2. Evidence

- **RIVUR (NEJM 2014):** 607 children with grade I-IV VUR, TMP-SMX vs placebo. CAP **halved recurrent UTI**, with the greatest benefit in BBD, but made **no difference to new scarring**.
- **Swedish Reflux Trial (2010):** in 1-2-year-olds with grade III-IV VUR, CAP and endoscopy reduced febrile UTI in **girls**; **boys** had few recurrences in any arm.

3. Treat BBD first: timed voiding, treat constipation, urotherapy, anticholinergics. BBD lowers resolution, increases breakthrough UTI and reduces endoscopic success.

4. Circumcision reduces UTI in infant boys.

5. Follow-up: RBUS every 6-12 months, VCUG every 12-24 months; monitor BP.

Guidelines

- **AUA (American Urological Association) 2010:** CAP is recommended for a child with BBD and VUR.
- **EAU/ESPU (European Association of Urology/European Society for Paediatric Urology):** high-risk children (grade IV-V, scars, BBD) start CAP; **intervene early** after breakthrough infection.

Indications for intervention: breakthrough febrile UTI, new scars, persistent grade IV-V VUR, poor CAP compliance, parental preference.

Case: 8-year-old boy with bilateral grade IV VUR

Assessment

- Exclude secondary VUR: PUV (urethra on VCUG), neurogenic bladder, BBD.
- DMSA, creatinine, BP and UTI history.
- At age 8 with bilateral grade IV, the chance of spontaneous resolution is **only about 10%**.

Options

1. **CAP with BBD management:** reasonable with no febrile UTI, no scars and a normal bladder. Boys past infancy have a lower UTI risk.
2. **Endoscopic injection (Deflux, dextranomer/hyaluronic acid)** by STING (subureteric) or HIT (hydrodistension-implantation). One injection succeeds in **about 63% of grade IV** (Elder 2006), more with repeats; day-case but less durable.
3. **Ureteral reimplantation:** the gold standard, with **95-98%** success.
 - **Cohen cross-trigonal** (intravesical): most popular and reliable; later ureteroscopic access is difficult.
 - **Lich-Gregoir** (extravesical): no ureterotomy, so less haematuria and spasm. **Bilateral repair risks transient retention** from pelvic plexus injury; nerve-sparing reduces it.
 - **RALUR (robotic-assisted laparoscopic ureteral reimplantation):** less pain and shorter stay; success about 85-95%.
 - Bilateral Cohen is ideal because one bladder opening corrects both sides.

Recommended plan

- If he has breakthrough febrile UTI, scars or loss of renal function: **bilateral reimplantation**. Deflux is an acceptable lower-morbidity alternative (about 60–70% success per session).
- If asymptomatic with no scars: CAP and BBD therapy; intervene if UTI or new scars occur.
- After surgery: CAP for about 3 months, RBUS at 4–6 weeks, long-term BP checks.

Exclude PUV, treat BBD -> UTI / scars / renal loss? Yes -> reimplant (or Deflux) No -> CAP + BBD therapy

Recent advances

- ceVUS (contrast-enhanced voiding urosonography) avoids radiation.
- Selective rather than universal CAP after RIVUR.

Key points to remember:

- IRS grade IV keeps papillary impressions; grade V loses them.
- RIVUR 2014: CAP halves recurrent UTI but does not change scarring.
- Deflux succeeds in about 63% of grade IV; open reimplantation in 95–98%.
- Bilateral extravesical reimplantation risks transient retention.

III-3a. Sacral neuromodulation

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation)** is continuous low-amplitude stimulation of the **S3 root** through a tined lead and an IPG (implantable pulse generator).
- It works by **afferent modulation**, not motor stimulation: somatic afferent input inhibits abnormal bladder reflexes at spinal and supraspinal levels, and in retention suppresses the sphincter "guarding reflex". Hence it helps **both overactivity and retention**.

Indications and FDA approvals

Indication	FDA (Food and Drug Administration) approval
Refractory UUI (urgency urinary incontinence)	1997
Urgency-frequency	1999
Non-obstructive retention , including Fowler's syndrome	1999
Faecal incontinence	2011

- Fowler's syndrome (young women with retention) responds best to SNM.

Devices

- InterStim and Axonics; newer devices are full-body MRI (magnetic resonance imaging) conditional.

Technique

- **Test phase:** either PNE (percutaneous nerve evaluation) for 3–7 days, or a **staged tined lead** for 1–2 weeks, which is more reliable.
- The needle enters the **S3 foramen** about 9 cm above the coccyx tip and 2 cm lateral.
- **S3 response:** bellows (pelvic floor contraction) plus big-toe plantar flexion.
- If symptoms improve by **50% or more**, the IPG is implanted in the upper buttock.

Outcomes

- **InSite (2015):** SNM beat standard medical therapy at 6 months, with benefit lasting 5 years.
- **ROSETTA (2016):** Botox 200 U gave slightly fewer UUI episodes than SNM at 6 months, but more UTIs (urinary tract infections) and self-catheterisation.

Complications

- IPG-site pain, lead migration, loss of effect, infection. Revision rates were historically about 30%.

Contraindications

- Failed test phase, inability to use the programmer, sacral anomalies.

Recent advances

- Smaller rechargeable generators; tibial nerve stimulation.

Diagram to draw: sacrum with the lead in S3 and the IPG in the buttock.

Key points to remember:

- Afferent modulation; helps overactivity and retention.
 - S3 response: bellows plus big-toe flexion.
 - Implant if the test gives at least 50% improvement.
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III-3b. Uses of buccal mucosa in urology

Type: SAQ (10 marks)

Introduction / history

- **BMG (buccal mucosal graft)** is the workhorse free graft for urethral reconstruction.
- **Humby 1941** first used it for hypospadias; **Bürger 1992** and **El-Kasaby 1993** popularised it for stricture.

Advantages

- Thick, **non-keratinised** epithelium that resists infection.
- **Thin, highly vascular lamina propria with a panlamina plexus** allows rapid imbibition and inosculation, so graft take is excellent.
- Hairless, used to a wet environment, easy to harvest, hidden scar, and usable in lichen sclerosus.

Harvest technique

- Nasotracheal intubation is preferred. Graft about 4–6 cm × 2–2.5 cm per cheek.
- Stay **1 cm from the lip commissure** and **avoid Stensen's duct**, which opens opposite the upper 2nd molar.
- Excise mucosa without muscle, achieve haemostasis, and **defat** the graft.
- The **lingual graft** from the tongue is the main alternative.

Donor-site complications

- Pain, perioral numbness, difficulty opening the mouth, Stensen's duct injury; mostly transient.

Uses

Area	Use
Anterior urethroplasty (main use)	Bulbar: dorsal onlay (Barbagli), ventral onlay, dorsal inlay (Asopa) , Kulkarni one-sided dorsal onlay, augmented anastomotic repair. Penile: dorsal inlay or onlay
Staged urethroplasty (Johanson, Bracka)	Lay open and graft; tubularise after about 6 months. For lichen sclerosus , failed hypospadias and long strictures
Hypospadias	Redo cases with no local skin
Ureter	Robotic BMG ureteroplasty for long proximal or mid strictures
Others	Bladder neck contracture, vaginoplasty, female urethral stricture

Outcomes

- Anterior BMG urethroplasty succeeds in about **85–90%**; dorsal and ventral onlay are similar.
- Results are poorer in penile, lichen sclerosus and redo cases.

Recent advances

- Tissue-engineered oral mucosa (investigational) and robotic BMG for the ureter and bladder neck.

Diagram to draw: cross-section of the bulbar urethra showing dorsal onlay, ventral onlay and Asopa inlay.

Key points to remember:

- Thin vascular lamina propria gives excellent take; avoid Stensen's duct.
 - Staged Johanson/Bracka repair for lichen sclerosus.
 - Anterior urethroplasty success is about 85-90%.
-

III-3c. HSG in neonates

Type: SAQ (10 marks)

HSG (hysterosalpingography) is not done in neonates, so this is taken as "USG (ultrasonography) in neonates".

Introduction

- First-line, radiation-free imaging, mainly for **ANH (antenatal hydronephrosis)** (about 1-2% of pregnancies).

Indications and timing

- Also for febrile UTI (urinary tract infection) and abdominal mass.
- **Avoid the first 48-72 hours**, because physiological oliguria underestimates dilatation.
- Scan **within 24-48 hours** if PUV (posterior urethral valves), bilateral severe hydronephrosis or a solitary kidney is suspected.
- If the first scan is normal, **repeat at 4-6 weeks**. Measure pelvic **APD (anteroposterior diameter)** on a transverse view.

Normal neonatal kidney

- **Prominent hypoechoic pyramids** are normal, not hydronephrosis; the cortex is as bright as the liver or brighter.

Grading of dilatation

- **SFU (Society for Fetal Urology) grades 0-4:** 0 none; 1 pelvis only; 2 pelvis and some calyces; 3 all calyces; 4 with parenchymal thinning.

UTD (urinary tract dilation) classification 2014, postnatal

Grade	Findings
P1 (low risk)	APD 10 to <15 mm or central calyceal dilatation
P2 (intermediate)	APD 15 mm or more , peripheral calyces or ureteric dilatation
P3 (high risk)	P2 plus abnormal parenchyma or abnormal bladder

Findings in specific conditions

Condition	USG findings
PUJO (pelviureteric junction obstruction)	Dilated pelvis and calyces, normal ureter
VUR (vesicoureteral reflux)	Fluctuating dilatation; a normal scan does not exclude VUR
PUV	Bilateral hydroureteronephrosis, thick-walled bladder , dilated posterior urethra. " Keyhole sign " is an antenatal USG sign; confirm postnatally with VCUG (voiding cystourethrogram)
MCDK (multicystic dysplastic kidney)	Non-communicating cysts

Follow-up algorithm

Postnatal USG (day 3-7; 24-48 h if PUV/bilateral)
Normal -> repeat at 4-6 wk; P1 -> USG at 1-6 mo
P2 -> USG at 1-3 mo, +/- VCUG
P3 -> USG at 1 mo, VCUG, antibiotic prophylaxis; urgent if PUV

Recent advances

- Contrast-enhanced voiding urosonography instead of VCUG.

Diagram to draw: SFU grades 0–4.

Key points to remember:

- Avoid the first 48–72 hours (oliguria), except for suspected PUV.
 - UTD 2014: P1 APD 10 to <15 mm; P2 15 mm or more.
 - A normal scan does not exclude VUR.
-

III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis?

Type: SAQ (10 marks)

Introduction

- Third-line treatment for ED (erectile dysfunction) failing tablets or injections; satisfaction is the highest of all ED treatments (**over 85-90%**).

Types

Type	Examples	Advantages	Disadvantages
Malleable	Genesis, Spectra, Tactra	Simple, cheap, no mechanical failure	Always rigid, hard to conceal
2-piece inflatable	Ambicor	No abdominal reservoir	Less flaccidity
3-piece inflatable	AMS 700 CX/LGX/CXR, Titan	Best rigidity and flaccidity	Cost, mechanical failure

Components of the 3-piece device

- Paired cylinders, a scrotal pump, and a reservoir in the space of Retzius or in an ectopic submuscular position after pelvic surgery.
- Mechanical survival is about 80-90% at 5 years.

Coatings

- **InhibiZone** (AMS) is impregnated with **rifampin and minocycline**.
- **Coloplast** hydrophilic coating absorbs antibiotic solution. Both roughly halve infection.

Biofilm

- **Definition:** a structured community of microorganisms in a self-produced **EPS (extracellular polymeric substance) matrix**, stuck to the implant.
- **Stages:** conditioning film, attachment, microcolonies, maturation, then dispersal causing clinical infection.
- **Organisms:** **coagulase-negative staphylococci (S. epidermidis)** commonest; also *S. aureus* and Gram-negatives.
- **Why antibiotics fail:** the matrix blocks penetration and the bacteria are dormant, so the MIC (minimum inhibitory concentration) may be **100-1000 times** higher.

Infection rates

- About **1-3% for first implants**; higher in revision surgery, poorly controlled diabetes and immunosuppression.

Prevention

- Chlorhexidine-alcohol preparation, intravenous antibiotics, **"no-touch" technique**, coated devices, short operating time.

Management of infection

- **Mulcahy salvage:** remove all components, wash out with a sequence of antiseptic solutions, then insert a new device at once. Success about **80-90%**.
- Not suitable with sepsis, necrosis or erosion; then explant.

Recent advances

- Tactra malleable device; ectopic reservoir placement after pelvic surgery.

Diagram to draw: 3-piece device in situ and the biofilm life cycle.

Key points to remember:

- Types: malleable, 2-piece and 3-piece inflatable.
 - Biofilm is a microbial community in an EPS matrix; *S. epidermidis* is commonest.
 - Infection about 1–3%; Mulcahy salvage succeeds in 80–90%.
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III-3e. Uretero-vaginal fistula

Type: SAQ (10 marks)

Definition

- **UVF (uretero-vaginal fistula):** abnormal ureter-vagina communication causing continuous vaginal leakage of urine.

Aetiology

- **Hysterectomy** is the commonest cause in developed countries (injury near the uterine artery or vaginal cuff).
- **Obstetric causes** (caesarean, obstructed labour) are common in developing countries; also cancer and radiation.

Clinical features

- **Continuous leakage with normal voiding**, because the bladder still fills from the other kidney.
- Usually appears **1-4 weeks after surgery**, because thermal or ischaemic injury takes time to necrose.

Investigations

- **Three-swab test** with methylene blue in the bladder: a blue swab means VVF (vesicovaginal fistula); a **wet but clear swab means UVF**.
- **CT (computed tomography) urography** shows contrast leak and hydronephrosis.
- **Cystoscopy and retrograde pyelogram** show the site; a stent is tried at the same sitting.
- **Exclude a coexisting VVF (about 10-15%)** and bilateral injury.

Management

1. Endourological (first line if continuity is preserved)

- **Retrograde double-J stent** if a guidewire crosses the defect. Otherwise PCN (percutaneous nephrostomy) and antegrade stent.
- Keep the stent for 6-8 weeks; many fistulae heal.

2. Surgical repair (failed stent, complete transection, ligation)

- Injuries are usually near the bladder, so do **ureteroneocystostomy: psoas hitch** bridges about 5 cm, **Boari flap** about 10-15 cm.
- **Early repair** (first few weeks) is preferred if stable without sepsis: outcomes match the traditional 3-6 month wait with less suffering.

Clear wet swab = UVF -> CT urography + cystoscopy (exclude VVF)
-> wire passes? Yes -> double-J 6-8 wk; No -> PCN + antegrade stent
-> fails -> reimplant + psoas hitch/Boari

Prevention

- Identify the ureter during hysterectomy; use prophylactic stents in complex cases.

Recent advances

- Robotic reimplantation; fluorescence to identify the ureter.

Diagram to draw: three-swab test; psoas hitch.

Key points to remember:

- Continuous leak with normal voiding; a wet clear swab means UVF.
 - Exclude a coexisting VVF (about 10–15%).
 - Stent if the wire passes; otherwise early reimplant.
-

III-3f. Prune belly syndrome

Type: SAQ (10 marks)

Definition

- **Eagle-Barrett** (triad) syndrome:
 1. Deficient abdominal wall muscles (wrinkled "prune" abdomen).
 2. Dilated urinary tract (ureters, bladder, prostatic urethra).
 3. Bilateral intra-abdominal undescended testes.
- Incidence about 1 in 30,000–50,000 live births; **about 95% are male**.

Pathogenesis

- **Mesodermal arrest** (6–10 weeks) affects abdominal wall, urinary tract muscle and gubernaculum.
- **Urethral obstruction theory**: transient fetal obstruction distends the bladder, stretching the abdominal wall and blocking testicular descent.
- Mostly sporadic; **CHRM3** mutations occur in familial cases.

Genitourinary features

- Kidneys: variable dysplasia, which decides the prognosis.
- Ureters: massively dilated and tortuous, with VUR (vesicoureteral reflux) in about 75%.
- Bladder: large, smooth-walled, poorly contracting.
- Prostatic hypoplasia with a dilated prostatic urethra. Testes are intra-abdominal, with infertility.

Extra-genitourinary features

- Pulmonary hypoplasia; cardiac defects in about 10%; orthopaedic problems in about 45%.

Woodard classification

Category	Features	Outlook
I	Oligohydramnios, pulmonary hypoplasia , renal dysplasia	Stillbirth or neonatal death
II	Full triad and significant uropathy, no pulmonary hypoplasia	Risk of urosepsis and CKD (chronic kidney disease)
III	Mild or incomplete features, stable renal function	Good

Management

- **Neonatal**: breathing support, creatinine (nadir predicts outcome), ultrasound, cautious VCUG (voiding cystourethrogram).
- **Most are managed conservatively** with prophylaxis, timed voiding and intermittent catheterisation if needed.
- **Drainage** for infection, rising creatinine or obstruction: **cutaneous vesicostomy** is preferred.
- **Orchiopexy**: transabdominal, **ideally by 6–12 months**; Fowler-Stephens if vessels are short.
- **Abdominoplasty**: Randolph, Ehrlich, or **Monfort** (most popular), often with orchiopexy.

Prognosis

- About one-third develop CKD; kidney transplant results are similar to other causes.

Females and incomplete forms

- Females (about 5%) have no testicular component. "Pseudo-prune belly" is an incomplete, milder form.

Recent advances

- Fetal shunting (benefit unproven); genetic testing.

Diagram to draw: VCUG with large bladder and dilated prostatic urethra.

Key points to remember:

- Triad: deficient abdominal wall, dilated urinary tract, intra-abdominal testes; about 95% male.
 - Most are managed conservatively; vesicostomy if drainage is needed.
 - Orchiopexy by 6-12 months, often with Monfort abdominoplasty.
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Rapid Revision Sheet - Paper III

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978: double-J stent.
- Alpha-blocker plus antimuscarinic best relieves stent symptoms.
- Encrusted stent: clear bladder coil, then ureter, then kidney.
- No routine stent after uncomplicated URS (AUA 2016, EAU 2024).

III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR. (LAQ)

- IRS grade IV keeps papillary impressions; grade V loses them.
- RIVUR 2014: CAP halves recurrent UTI but does not change scarring.
- Deflux succeeds in about 63% of grade IV; open reimplantation in 95–98%.
- Bilateral extravesical reimplantation risks transient retention.

III-3a. Sacral neuromodulation (SAQ)

- Afferent modulation; helps overactivity and retention.
- S3 response: bellows plus big-toe flexion.
- Implant if the test gives at least 50% improvement.

III-3b. Uses of buccal mucosa in urology (SAQ)

- Thin vascular lamina propria gives excellent take; avoid Stensen's duct.
- Staged Johanson/Bracka repair for lichen sclerosus.
- Anterior urethroplasty success is about 85–90%.

III-3c. HSG in neonates (SAQ)

- Avoid the first 48–72 hours (oliguria), except for suspected PUV.
- UTD 2014: P1 APD 10 to <15 mm; P2 15 mm or more.
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III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis? (SAQ)

- Types: malleable, 2-piece and 3-piece inflatable.
- Biofilm is a microbial community in an EPS matrix; *S. epidermidis* is commonest.
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III-3e. Uretero-vaginal fistula (SAQ)

- Continuous leak with normal voiding; a wet clear swab means UVF.
- Exclude a coexisting VVF (about 10–15%).
- Stent if the wire passes; otherwise early reimplant.

III-3f. Prune belly syndrome (SAQ)

- Triad: deficient abdominal wall, dilated urinary tract, intra-abdominal testes; about 95% male.
- Most are managed conservatively; vesicostomy if drainage is needed.
- Orchiopexy by 6–12 months, often with Monfort abdominoplasty.